CAPSTONE SYSTEM ANALYSIS & DESIGN REPORT

ELEMENTAL: Adaptive Korean Language Learning

Team: Abdujalil & Planning Team **Focus:** TOPIK Practice & Adaptive Learning

1. Project Plan & Objectives

Independent Korean language learners frequently rely on fragmented materials—ranging from static note sheets to standalone video playlists. While informative, these disparate resources lack a structured learning continuum and fail to deliver objective feedback on mastery. Completing a reading material does not ensure comprehension.

ELEMENTAL addresses these challenges by consolidating lessons, quizzes, immediate feedback, and historical progress metrics into a single unified platform. Its central innovation is a *rule-based adaptive difficulty mechanism* that dynamic adjusts practice challenge levels based on evaluated student accuracy.

1.2 Scope and User Roles

Scope Area	Inclusions & System Responsibilities
User Access	Role-based authentication for Learners and Content Administrators.
Content Management	Structured hierarchy: TOPIK Levels → Learning Topics → Lessons → Quizzes.
Adaptive Practice	Rule-based difficulty tiering, score calculation, and progress dashboards.
Out of Scope	Payment processing, live tutoring, AI generation, and software deployment.

1.3 System Business Rules

> **BR-1 (Prerequisite Access):** A learner can access advanced topic quizzes only after viewing core topic lessons.

> **BR-2 (Adaptive Advancement):** Achieving ≥80% accuracy across two consecutive attempts at the current tier advances the learner to the next difficulty tier.

> **BR-3 (Foundational Reinforcement):** Scoring <50% on a quiz triggers reinforcement tier routing and targets fundamental review.

> **BR-4 (Content Integrity):** Quizzes cannot be published without valid answer keys assigned to every question.

> **Summary:** The project plan defines the problem domain, target users, functional scope boundaries, and core business rules that govern the adaptive algorithms and relational database models.

SYSTEM CONSISTENCY & CONCLUSION

Consistency Matrix & Project Summary

2. Cross-Model Consistency Matrix

To ensure structural alignment across design artifacts, every functional use case directly maps to its corresponding business process model, primary relational database entities, and user interface layouts.

Use Case	Business Process	Primary Relational Tables	Target UI View
UC-01: Register & Sign In	BPM 1 / BPM 2	`LEARNER`, `ADMINISTRATOR`	Auth Portal
UC-02: Browse & Study Lessons	BPM 1	`LEVEL`, `TOPIC`, `LESSON`	Lesson Catalog
UC-03: Take Adaptive Quiz	BPM 3	`QUIZ`, `QUESTION`, `QUIZ_ATTEMPT`, `LEARNER_PROGRESS`	Quiz Execution
UC-04: Manage Content	BPM 2	`TOPIC`, `LESSON`, `QUESTION`, `DIFFICULTY_TIER`	Admin Studio
UC-05: View Dashboard	BPM 3	`LEARNER_PROGRESS`, `QUIZ_ATTEMPT`	Progress View

3. Project Conclusion

The **ELEMENTAL** capstone analysis demonstrates how a modern, adaptive language learning platform can be rigorously modeled prior to implementation. By defining exact specifications, modeling operational process workflows with BPMN, and establishing normal relational data structures via Chen and IE Crow's Foot diagrams, the project provides a robust design foundation.

The rule-based adaptive engine ensures learners receive targeted practice matched to their evolving capabilities, resolving the core problem of static, unguided self-study.

> **Final Note:** All entity names, actor definitions, and data types remain consistent across every diagram and dictionary table, ensuring the system analysis ready for software engineering execution.